

Hacker Rank Day 2

Task 1

The provided code stub reads two integers, a and b , from STDIN.

Add logic to print two lines. The first line should contain the result of integer division, $a // b$. The second line should contain the result of float division, a / b .

No rounding or formatting is necessary.

Example

$$a = 3$$

$$b = 5$$

- The result of the integer division $3 // 5 = 0$.
- The result of the float division is $3 / 5 = 0.6$.

Print:

0

0.6

Input Format

The first line contains the first integer, a .

The second line contains the second integer, b .

Output Format

Print the two lines as described above.

Sample Input 0

4

3

Sample Output 0

1

1.333333333333

Leaderboard

Discussions

Editor

Check the [Tutorial](#) tab to know learn about division operators.

Task

The provided code stub reads two integers, *a* and *b*, from STDIN.

Add logic to print two lines. The first line should contain the result of integer division, *a* // *b*. The second line should contain the result of float division, *a* / *b*.

No rounding or formatting is necessary.

Example

a = 3

b = 5

- The result of the integer division *3* // *5* = 0.
- The result of the float division *3* / *5* = 0.6.

Print:

0

Change Theme

Language

Pypy 3

```
1 if __name__ == '__main__':
2     a = int(input())
3     b = int(input())
4
5     print(a//b)
6     print(a/b)
7
```

Leaderboard

Discussions

Editorial

Tutorial

HackerRank

PreparePythonIntroductionPython: Division

Exit Full Screen View

Check the [Tutorial](#) tab to know learn about division operators.

Task

The provided code stub reads two integers, a and b , from STDIN.
Add logic to print two lines. The first line should contain the result of integer division, $a // b$. The second line should contain the result of float division, a / b .
No rounding or formatting is necessary.

Example

$a = 3$
 $b = 5$

- The result of the integer division $3 // 5 = 0$.
- The result of the float division is $3 / 5 = 0.6$.

Print:

```
0
0.6
```

Input Format

The first line contains the first integer, a .
The second line contains the second integer, b .

Output Format

Print the two lines as described above.

You have earned 10.00 points!

You are now 35 points away from the 2nd star for your python badge.

0%35/70

Congratulations

You solved this challenge. Would you like to challenge your friends?

[Next Challenge](#)

Test case 0

Test case 1

Compiler Message

Success

Input (stdin)

Expected Output

Download

Download

21:06:33

24-06-2023

The first and only line contains the integer, .

Constraints

$$1 \leq n \leq 20$$

Output Format

Print n lines, one corresponding to each i .

Sample Input 0

5

Sample Output 0

0

1

4

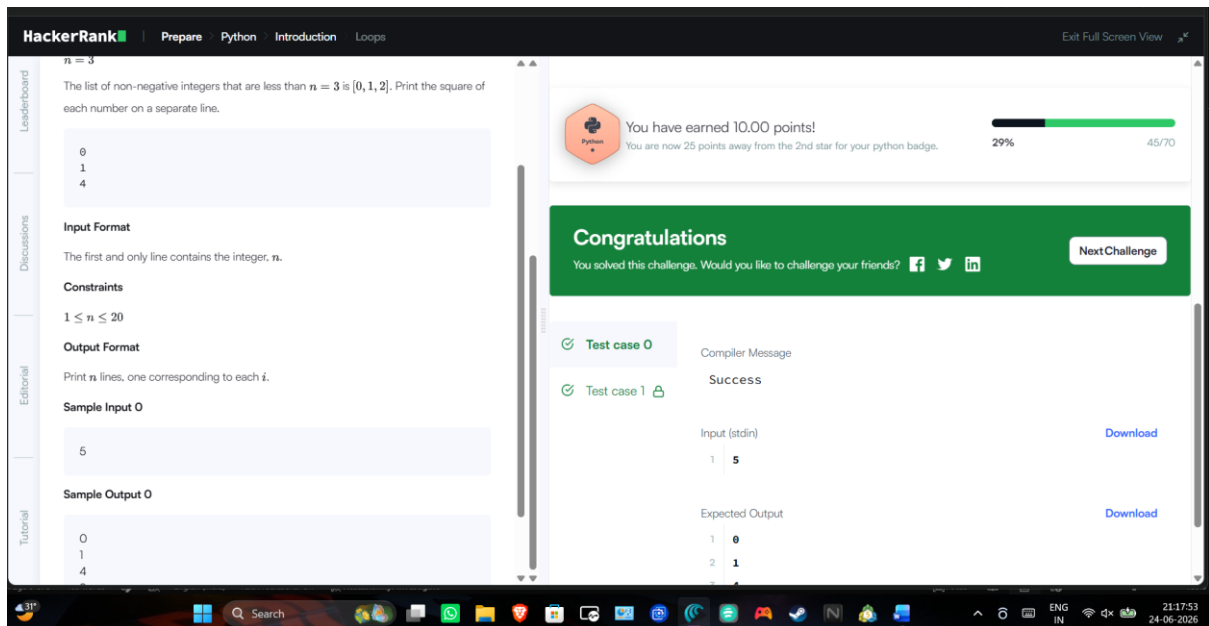
9

16

The screenshot shows the HackerRank Python editor interface. On the left, a sidebar contains links for Leaderboard, Discussions, Editorial, and Tutorial. The main content area on the left displays the problem description: "The list of non-negative integers that are less than $n = 3$ is $[0, 1, 2]$. Print the square of each number on a separate line." Below this, it shows the input format, constraints ($1 \leq n \leq 20$), output format (Print n lines, one corresponding to each i), sample input 0 (5), and sample output 0 (0, 1, 4). The main code editor on the right shows a Python script that reads an integer n and prints the squares of integers from 0 to $n-1$. The script is as follows:

```
1 if __name__ == '__main__':
2     n = int(input())
3     for i in range(0,n):
4         print(i*i)
```

The status bar at the bottom right of the editor indicates "Line: 4 Col: 18". At the bottom of the interface, there are buttons for "Upload Code as File", "Test against custom input", "Run Code", and "Submit Code". The Windows taskbar is visible at the very bottom of the image.



Task 3

An extra day is added to the calendar almost every four years as February 29, and the day is called a *leap day*. It corrects the calendar for the fact that our planet takes approximately 365.25 days to orbit the sun. A leap year contains a leap day.

In the Gregorian calendar, three conditions are used to identify leap years:

- The year can be evenly divided by 4, is a leap year, unless:
 - The year can be evenly divided by 100, it is NOT a leap year, unless:
 - The year is also evenly divisible by 400. Then it is a leap year.

This means that in the Gregorian calendar, the years 2000 and 2400 are leap years, while 1800, 1900, 2100, 2200, 2300 and 2500 are NOT leap years.

Given a year, determine whether it is a leap year. If it is a leap year, return the Boolean True, otherwise return False.

Note that the code stub provided reads from STDIN and passes arguments to the `is_leap` function. It is only necessary to complete the `is_leap` function.

Input Format

Read , the year to test.

Constraints

$$1900 \leq year \leq 10^5$$

Output Format

The function must return a Boolean value (True/False). Output is handled by the provided code stub.

Sample Input 0

1990

Sample Output 0

False

Explanation 0

1990 is not a multiple of 4 hence it's not a leap year.

The screenshot shows the HackerRank Python editor interface. On the left, the problem description for 'is_leap' is visible, detailing the Gregorian calendar rules for leap years. The main editor area contains the following Python code:

```
def is_leap(year):  
    leap = False  
  
    # Write your logic here  
    if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):  
        leap = True  
  
    return leap  
  
year = int(input())  
print(is_leap(year))
```

The code is currently at Line 5, Column 55. The 'Run Code' and 'Submit Code' buttons are visible at the bottom right of the editor.

The screenshot shows the HackerRank Python editor interface after the code has been executed. The test results are displayed on the right side of the editor:

- Test case 0:** Success
- Test case 1:** Success
- Test case 2:** Success
- Test case 3:** Success

A congratulatory message is shown: "You have earned 10.00 points! You are now 15 points away from the 2nd star for your python badge." The progress bar indicates 57% completion (55/70 points). A "Next Challenge" button is available. A "Hidden Test Case" is also mentioned, stating "Unlock this test case for 5 hacks."

Task 4

The included code stub will read an integer, n , from STDIN.

Without using any string methods, try to print the following:

Note that `""` represents the consecutive values in between.

Example

Print the string 12345.

Input Format

The first line contains an integer n .

Constraints

$$1 \leq n \leq 150$$

Output Format

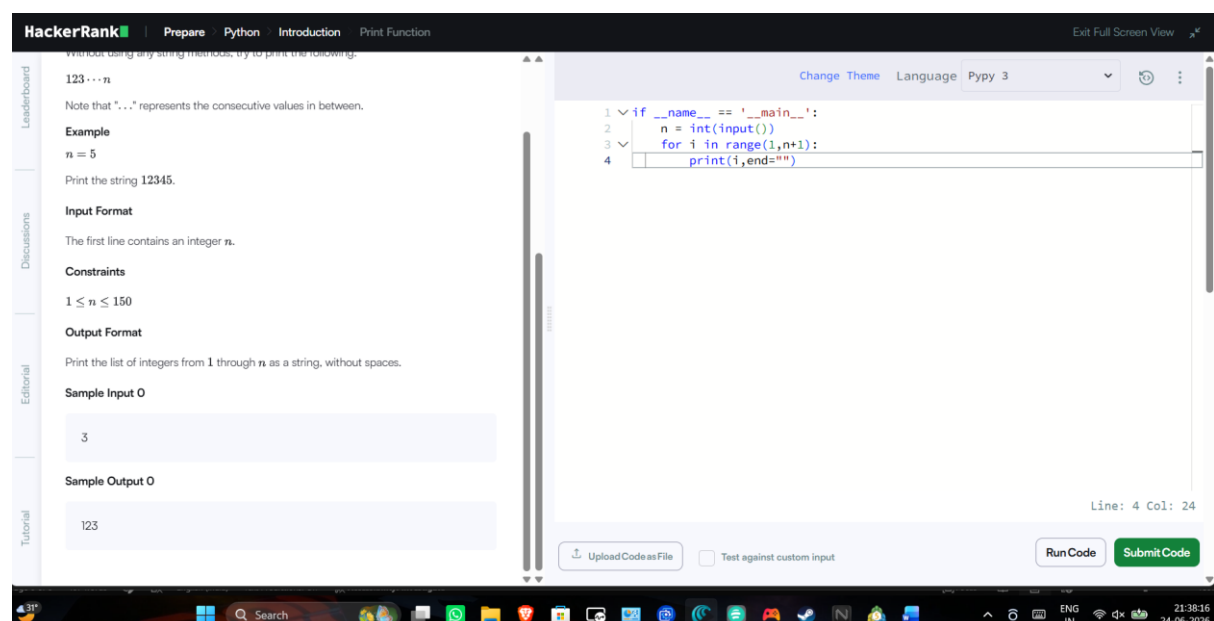
Print the list of integers from 1 through n as a string, without spaces.

Sample Input 0

3

Sample Output 0

123



$123 \cdots n$

Example

Input Format

Constraints

$$1 \leq n \leq 150$$

Output Format

Print the list of integers from 1 through n as a string, without spaces.

Sample Input 0

3

Sample Output 0

123

You have earned 20.00 points!

You are now 35 points away from the 3rd star for your python badge.

3704

75/110

You solved this challenge. Would you like to challenge your friends?

Next Challenge

✔ Test case 0

Compiler Message

Success

✔ Test case 1

Test case 2

Input (stdin)

1 | 3

Download

Expected Output

1 123

[Download](#)

Task 5

Consider a list (`list = []`). You can perform the following commands:

1. `insert i e`: Insert integer e at position i .
2. `print`: Print the list.
3. `remove e`: Delete the first occurrence of integer e .
4. `append e`: Insert integer e at the end of the list.
5. `sort`: Sort the list.
6. `pop`: Pop the last element from the list.
7. `reverse`: Reverse the list.

Initialize your list and read in the value of n followed by n lines of commands where each command will be of the 7 types listed above. Iterate through each command in order and perform the corresponding operation on your list.

Example

$N = 4$

`append 1`

`append 2`

`insert 1 3`

`print`

- `append 1`: Append 1 to the list, $arr = [1]$.
- `append 2`: Append 2 to the list, $arr = [1, 2]$.
- `insert 1 3`: Insert 3 at index 1, $arr = [1, 3, 2]$.
- `print`: Print the array.
- Output:

[1, 3, 2]

Input Format

The first line contains an integer, n , denoting the number of commands.

Each line of the subsequent lines contains one of the commands described above.

Constraints

- The elements added to the list must be *integers*.

Output Format

For each command of type `print`, print the list on a new line.

Sample Input 0

12

insert 0 5

insert 1 10

insert 0 6

print

remove 6

append 9

append 1

sort

print

pop

reverse

print

Sample Output 0

[6, 5, 10]

[1, 5, 9, 10]

[9, 5, 1]

HackerRank | Prepare | Python | Basic Data Types | Lists | Exit Full Screen View

• The elements added to the list must be integers.

Output Format
For each command of type print, print the list on a new line.

Sample Input 0

```
12
insert 0 5
insert 1 10
insert 0 6
print
remove 6
append 9
append 1
sort
print
pop
reverse
print
```

Sample Output 0

```
[6, 5, 10]
[1, 5, 9, 10]
[9, 5, 1]
```

Change Theme | Language: Pypy 3

```
1 if __name__ == '__main__':
2     N = int(input())
3     list = []
4     for _ in range(N):
5         com = input().split()
6
7         if com[0] == "insert":
8             list.insert(int(com[1]),int(com[2]))
9         elif com[0] == "print":
10            print(list)
11        elif com[0] == "remove":
12            list.remove(int(com[1]))
13        elif com[0] == "append":
14            list.append(int(com[1]))
15        elif com[0] == "sort":
16            list.sort()
17        elif com[0] == "pop":
18            list.pop()
19        elif com[0] == "reverse":
20            list.reverse()
21
22
```

Line: 18 Col: 22

Upload Code as File | Test against custom input | Run Code | Submit Code

- The elements added to the list must be integers.

Output Format

For each command of type `print`, print the list on a new line.

Sample Input 0

```
12
insert 0 5
insert 1 10
insert 0 6
print
remove 6
append 9
append 1
sort
print
pop
reverse
print
```

Sample Output 0

- [6, 5, 10]
- [1, 5, 9, 10]
- [9, 5, 1]



You have earned 10.00 points!

You are now 25 points away from the 3rd star for your python badge.

38%

85/110

Congratulations

You solved this challenge. Would you like to challenge your friends?



Next Challenge

✔ Test case 0

✔ Test case 1 

```
3 insert 1 10
4 insert 0 6
5 print
6 remove 6
7 append 9
8 append 1
9 sort
10 print
11 pop
12 reverse
13 print
```